

PhyST-Net: An Exogenous-Aware Spatio-Temporal Graph Transformer for Domain-Constrained Network Forecasting

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Abstract—Spatio-temporal forecasting is a core task for graph-structured networks such as power grids, traffic systems, and industrial sensor clusters, where accurate node-level prediction requires joint modeling of historical trends, external drivers, and operational constraints. Existing forecasting models often treat historical covariates, future exogenous variables, and domain priors as isolated inputs, limiting their ability to capture the interplay between temporal continuity, dynamic spatial topology, and physical feasibility. This paper proposes PhyST-Net, an exogenous-aware spatio-temporal graph Transformer designed as a general framework for domain-constrained network forecasting. PhyST-Net formulates prediction as a multi-modal graph learning task with historical target states, historical exogenous variables (ExG), future ExG, and graph metadata as distinct interfaces. The framework employs: (1) Adaptive Gaussian-soft-windowed Patching (AGSP) for continuous temporal tokenization; (2) a Relational Spatio-Temporal Manifold Fusion (R-STMF) graph adapter for multi-scale spatial dependency modeling; (3) KAN-enhanced Adaptive Conditioning (K-AC) for non-linear exogenous modulation; and (4) a Domain-Informed Dynamic Capping Head (DI-DCH) for constraint-aware decoding. While the framework is general, we validate its effectiveness on the challenging task of wind power forecasting using the SD-WPF, HoT, and Kelmarsh datasets. Results show that PhyST-Net consistently improves forecasting accuracy and feasibility across multiple horizons, demonstrating the value of explicit exogenous awareness and domain-informed architectural inductive biases.

Index Terms—Spatio-temporal forecasting, graph Transformers, exogenous variables, domain-constrained machine learning, network forecasting.

I. INTRODUCTION

SPATIO-TEMPORAL forecasting aims to predict future node states in graph-structured systems where temporal dynamics and spatial interactions evolve jointly, such as transportation networks, power grids, and industrial sensor clusters. In these applications, node-level trajectories are rarely self-driven; they are shaped by historical operating conditions, future external drivers, and domain-specific feasibility constraints. As illustrated in Fig. 1, complex network signals are often composed of shared macro-trends driven by exogenous forcing and high-frequency variations unique to individual nodes. A practical model must therefore learn continuous temporal representations, infer dynamic graph relations, and integrate future exogenous variables (ExG) without data leakage, all while ensuring predictions remain compatible with operational operating regimes.

Existing methods often leave the complete data flow implicit, treating historical covariates, future drivers, and physical

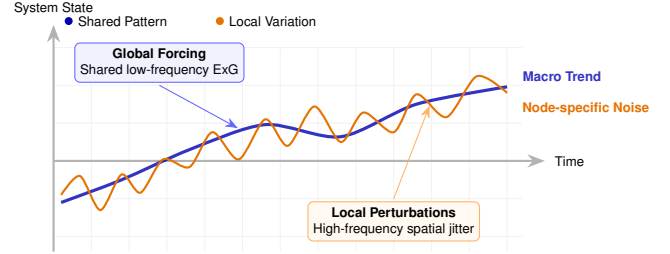


Fig. 1. Multi-scale spatio-temporal processes. General network signals are composed of shared macro-trends driven by exogenous forcing and high-frequency variations unique to individual nodes. PhyST-Net separates these components through dual-branch graph reasoning.

priors as loosely coupled inputs. Standard Transformers employ rigid patching that may split regime transitions across artificial boundaries. Most graph neural networks conflate multi-scale interactions into a single operator, failing to distinguish between local dependencies and global environmental fronts. Furthermore, physical feasibility is typically enforced through post-processing, which detaches the forecast from domain-specific operating envelopes. These limitations necessitate an explicit exogenous-aware framework where the forecasting contract—historical states, observed context, future drivers, and domain constraints—is exposed as a first-class architectural principle.

This paper presents PhyST-Net, an ExG-aware Spatio-Temporal Graph Transformer designed as a general framework for domain-constrained network forecasting. To address the multi-scale and multi-modal nature of network signals, we introduce: (1) **Adaptive Gaussian-soft-windowed Patching (AGSP)**, a differentiable temporal partitioning mechanism that transforms discrete observations into learnable spectral-temporal densities, thereby recovering continuous event dynamics often distorted by rigid patching; (2) **Relational Spatio-Temporal Manifold Fusion (R-STMF)**, a modular graph adapter that decomposes latent node dynamics into multi-scale spatial manifolds, using structural pruning to filter spurious correlations and recover physical interaction streamlines; (3) **KAN-enhanced Adaptive Conditioning (K-AC)**, which leverages Kolmogorov-Arnold spline edge functions to perform audit-ready non-linear modulation of graph-enhanced states using future exogenous drivers; and (4) a **Domain-Informed Dynamic Capping Head (DI-DCH)** that enforces physical feasibility directly within the decoding path by mapping high-dimensional representations into a dynamic, domain-specific feasible set.

By explicitly separating the data flow into distinguishable pathways—historical context, relational reasoning, and ex-

ogenous conditioning—PhyST-Net maintains a rigorous forecasting contract. This interface-centric design prevents the conflation of historical and future drivers, ensuring that the architecture can be seamlessly instantiated across diverse domains, such as traffic networks, power grids, and distributed industrial clusters, where predictions must be both data-driven and structurally sound.

A. Contributions

The main contributions are summarized as follows:

- **A general ExG-aware graph-forecasting formulation.** We define network forecasting as a multi-modal task with historical states, exogenous variables, and domain constraints exposed as distinguishable pathways, supporting broad framework generalizability.
- **A continuous temporal tokenization module.** We introduce AGSP, which replaces rigid patches with learnable Gaussian soft-windows to preserve event continuity and reduce boundary artifacts in non-stationary sequences.
- **A modular graph adapter for mixed spatial mechanisms.** We develop R-STMF to separate multi-scale spatial dependencies before relational propagation, using a modular contract that allows for easy substitution of alternative graph predictors.
- **A future-ExG-conditioned decoding path.** We combine K-AC and DI-DCH to inject forecast-horizon drivers and enforce feasibility constraints inside the model, linking latent representation learning directly to operational requirements.

The effectiveness of PhyST-Net is validated through extensive experiments on real-world datasets across diverse scales and operational regimes, demonstrating its robustness as a general-purpose network forecasting framework.

II. RELATED WORK

This section reviews representative research lines related to PhyST-Net. Rather than listing models chronologically, we organize prior work by the modeling challenges that are central to ExG-aware spatio-temporal forecasting: temporal dependency modeling, graph dependency learning, exogenous-variable fusion, and constraint-aware forecasting.

A. Temporal Forecasting and Transformer Models

Classical forecasting methods, including autoregressive models, state-space models, and exponential smoothing, remain valuable when the target process is low-dimensional and approximately stationary. Their assumptions, however, become restrictive for graph-structured systems where node dynamics are nonlinear and coupled with external drivers. Deep recurrent and convolutional architectures improve nonlinear sequence modeling by learning hidden states or local temporal filters, but long-horizon forecasting remains difficult because relevant information may span many time steps and because error accumulation can become severe in multi-step prediction.

Transformer-based forecasters address long-range dependency modeling through attention and have become a major

family of deep time-series models. Variants such as PatchTST-style patching and inverted-dimension attention reduce computational cost and improve representation learning for multivariate sequences [1], [5], [6]. Nevertheless, temporal patching is usually implemented through fixed windows. In non-stationary sequences, regime transitions and rapid transitions do not necessarily align with fixed patch boundaries. PhyST-Net follows the Transformer forecasting direction but introduces AGSP to make temporal tokenization learnable and continuous.

B. Spatio-Temporal Graph Forecasting

Spatio-temporal graph neural networks (STGNNs) model node interactions through graph convolution, diffusion propagation, graph attention, or learned adjacency matrices [9], [10], [14]. These methods are effective when cross-node dependencies are important, as in traffic networks, sensor systems, and large-scale networks. Adaptive graph learning further relaxes the assumption that the observed topology is complete or stationary. It can infer latent relations from data and adjust the propagation pattern to changing operating conditions.

The limitation is that many graph modules apply a homogeneous propagation rule to all latent states. In physical graph systems, spatial dependence may contain multiple mechanisms. For example, nodes can be related through local interactions, environmental influences, and shared exogenous systems. A single learned adjacency may capture correlation but may not distinguish these mechanisms. R-STMF addresses this issue by decomposing temporal states into local-variation and macro-trend components and applying sparse relational propagation before fusion. This design is intended to preserve the benefit of adaptive graph learning while making the spatial pathway more interpretable at the representation level.

C. Exogenous-Variable-Aware Forecasting

Practical forecasting tasks often depend on exogenous variables such as weather forecasts, calendars, control schedules, market signals, and planned events. In spatio-temporal node prediction, external measurements and exogenous variables are essential because the target signal depends on environmental factors and node operating states. Existing forecasting models incorporate external information through concatenation, attention, encoder-decoder fusion, or graph-enhanced fusion [26], [27]. These mechanisms can improve accuracy when covariates are reliable and well aligned.

A key issue is that historical ExG and future ExG play different roles. Historical ExG explains the recent trajectory and helps estimate the current latent state, while future ExG provides horizon-level information that should modulate the forecast. Merging both into a single feature tensor can blur this temporal distinction and make it difficult to reason about data availability at inference time. PhyST-Net therefore separates historical ExG from future ExG in the problem definition. K-AC uses the future-ExG pathway to generate latent modulation parameters, allowing forecast-horizon drivers to condition graph-enhanced states without being treated as ordinary historical observations.

D. Interpretable Conditional Modeling

Interpretability is increasingly important in operational forecasting because users often need to understand not only the predicted value but also the drivers that shape the forecast. Attention weights, feature attribution, and post-hoc explanation methods provide useful diagnostics, but they do not always correspond to the actual computational pathway used by the model. Conditional forecasting modules are especially difficult to interpret when future drivers are fused through dense layers, because the contribution of a specific driver becomes distributed across many hidden units.

KAN-style representations offer a different form of interpretability by modeling transformations through learnable univariate functions. When a conditioning network is written as an additive sum of feature-wise edge functions, the influence of each input variable can be inspected through its learned basis coefficients and derivatives. PhyST-Net uses this property in K-AC: future ExG is mapped to scale and shift parameters through spline-based functions, making it possible to analyze how each exogenous driver modulates the latent graph state. This does not imply causal identification, but it provides an architecture-aligned explanation mechanism that is more transparent than generic feature concatenation.

E. Scalable Spatio-Temporal Forecasting

Scalability is a recurring issue in graph forecasting. Dense attention over all time steps and all node pairs can become expensive when both the look-back length and graph size are large. Existing approaches reduce complexity through temporal downsampling, patching, sparse attention, graph sampling, diffusion truncation, or low-rank structure. These techniques improve efficiency but may also remove useful temporal or spatial information if the reduction is too aggressive.

PhyST-Net addresses scalability at two levels. AGSP reduces the temporal dimension from raw historical length to a smaller number of continuous tokens, preserving soft event boundaries while lowering the attention cost. R-STMF uses sparse top- K graph propagation so that node interaction cost can be controlled by the retained neighborhood size rather than by the full graph size. The complexity analysis in Section IV-H makes these design choices explicit and connects the architecture to practical deployment constraints.

F. Domain-Informed and Constraint-Aware Forecasting

Domain-informed learning introduces domain knowledge into neural models through structural priors, loss penalties, differentiable simulators, or post-processing rules. In network forecasting, domain constraints are especially relevant because the target signal is bounded by operational limits and node regimes depend on external thresholds. Forecasts that violate these constraints can harm downstream decisions even when aggregate error metrics appear acceptable.

Existing constraint-aware approaches often use physics-guided losses [24] or post-hoc clipping [20], [25]. Loss penalties are flexible but do not guarantee feasible outputs. Post-hoc clipping guarantees bounded values but can create a mismatch between the function learned during training

and the projection used at inference. DI-DCH places the feasibility operation inside the forecasting head. The head first shapes predictions toward a dynamic capacity envelope and then applies a final projection. This makes the constraint pathway visible in the computational graph while retaining compatibility with standard supervised forecasting losses.

G. Position of This Work

PhyST-Net is closest to methods that combine Transformer forecasting, adaptive graph learning, and exogenous conditioning. Its difference lies in the explicit organization of the forecasting interface. Instead of presenting network forecasting only as a domain-specific architecture, the paper formulates it as ExG-aware graph-structured node prediction and then instantiates domain-specific modules where physical priors are meaningful. This position allows the method to be evaluated as a general-purpose model while also providing a reusable template for other graph forecasting domains with future external drivers and output constraints.

III. PROBLEM DEFINITION

We formulate the task as ExG-aware spatio-temporal graph forecasting. Let $G = (V, E, A)$ denote a graph, where $V = \{v_1, \dots, v_N\}$ is the node set, E is the edge set, and $A \in \mathbb{R}^{N \times N}$ is an adjacency or relation matrix. A node can represent a network element, road sensor, grid bus, monitoring station, or industrial device depending on the application. The graph relation A may be physical, geographic, statistical, learned, or a combination of these sources.

At forecasting origin t , the historical target tensor is

$$\mathbf{X}_{t-T+1:t} \in \mathbb{R}^{T \times N \times D_x}, \quad (1)$$

where T is the look-back length, N is the number of nodes, and D_x is the target-state dimension. The future prediction target is

$$\mathbf{Y}_{t+1:t+H} \in \mathbb{R}^{H \times N \times D_y}, \quad (2)$$

where H is the forecast horizon and D_y is the output dimension.

We distinguish two types of exogenous variables. Historical ExG is available over the look-back window:

$$\mathbf{E}_{t-T+1:t}^h \in \mathbb{R}^{T \times D_h}, \quad (3)$$

and describes external or contextual conditions observed together with the target history. Future ExG is available over the forecast horizon:

$$\mathbf{E}_{t+1:t+H}^f \in \mathbb{R}^{H \times D_f}, \quad (4)$$

and represents known or forecastable future drivers. This separation is important because historical ExG estimates the current latent state, whereas future ExG conditions the unknown horizon. In spatio-temporal node prediction, historical ExG may include external variables and time features, while future ExG may include exogenous forecasts.

The formulation assumes an operational forecasting setting. At time t , the model has access to historical target states and historical ExG up to t , and it has access to future ExG for

$t + 1$ to $t + H$ only if those variables are known, scheduled, or forecast by an external source. The future target values are never used as inputs. This assumption is important for preventing leakage: variables that are measured only after the target horizon should not be included in $\mathbf{E}_{t+1:t+H}^f$. If an application lacks reliable future ExG, the same formulation can be used by setting $\mathbf{E}^f$ to calendar features, scenario forecasts, or a learned uncertainty representation, but the resulting claim should be stated as a different forecasting setting.

The graph relation is also allowed to be partially observed. In some systems, A is a physical adjacency matrix; in others, it is a distance kernel, a correlation prior, or an empty prior that is refined by adaptive learning. We therefore separate A from M_G . The matrix A specifies direct relation priors used by graph propagation, while M_G collects auxiliary metadata such as node coordinates, node attributes, or environmental descriptors. This separation lets the same method support both fixed-topology and adaptive-topology forecasting.

The forecasting model is a parameterized function f_θ :

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_\theta(\mathbf{X}_{t-T+1:t}, \mathbf{E}_{t-T+1:t}^h, \mathbf{E}_{t+1:t+H}^f, A, M_G), \quad (5)$$

where M_G denotes graph metadata such as coordinates, node attributes, physical distances, or topology priors. The model should satisfy three requirements. First, it should extract temporal representations from non-stationary node histories. Second, it should propagate information across graph nodes without assuming that all spatial dependencies share one mechanism. Third, it should use future ExG only through variables available at inference time.

When domain constraints are available, the model can be decomposed into an unconstrained spatio-temporal backbone $\mathcal{H}_\theta$ and a constraint operator $\mathcal{C}_\Omega$:

$$\hat{\mathbf{Y}}_{t+1:t+H} = \mathcal{C}_\Omega\left(\mathcal{H}_\theta\left(\mathbf{X}_{t-T+1:t}, \mathbf{E}_{t-T+1:t}^h, \mathbf{E}_{t+1:t+H}^f, A, M_G\right)\right) \quad (6)$$

where Ω denotes domain knowledge. In spatio-temporal node prediction, Ω includes non-negativity, capacity limits, and regime feasibility. In other domains, Ω may encode speed limits, conservation constraints, bounded sensor ranges, or operational safety limits.

Given a training set $\mathcal{D}$, the supervised objective is

$$\min_{\theta} \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G, \mathbf{Y}) \in \mathcal{D}} \ell(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda_\Omega \mathcal{R}_\Omega(\hat{\mathbf{Y}}, \mathbf{Y}), \quad (7)$$

where $\ell(\cdot)$ is a forecasting loss, $\mathcal{R}_\Omega$ is an optional constraint-aware regularizer, and λ_Ω controls its strength. The formulation does not require a specific graph operator; any module that maps temporal node states to graph-enhanced node states can be used as the graph adapter.

IV. METHOD

PhyST-Net is an ExG-aware spatio-temporal graph Transformer designed for network-structured node forecasting with known external drivers and domain constraints. Given the inputs defined in Section III, the model follows the pipeline in Fig. 2. Historical target states and historical ExG are first

embedded into node-level temporal representations. AGSP converts the temporal history into continuous tokens. A Transformer backbone learns temporal dependencies. R-STMF serves as a graph adapter that maps temporal node states to graph-enhanced node states. K-AC injects future ExG into the latent representation. DI-DCH finally decodes the conditioned states into feasible forecasts respecting domain-specific bounds.

A. Overall Framework

For a mini-batch, let $\mathbf{X} \in \mathbb{R}^{B \times T \times N \times D_x}$ denote historical target states, $\mathbf{E}^h \in \mathbb{R}^{B \times T \times D_h}$ denote historical ExG, and $\mathbf{E}^f \in \mathbb{R}^{B \times H \times D_f}$ denote future ExG. The model produces $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times N \times D_y}$. The computation can be written as

$$Z = \Phi_{patch}(\mathbf{X}, \mathbf{E}^h, M_G), \quad (8)$$

$$H_T = \Phi_{temp}(Z), \quad H_G = \Phi_{graph}(H_T, A, M_G), \quad (9)$$

$$H_F = \Phi_{exg}(H_G, \mathbf{E}^f), \quad \hat{\mathbf{Y}} = \Phi_{head}(H_F, \mathbf{E}^f, \Omega), \quad (10)$$

where Φ_{patch} is AGSP, Φ_{temp} is the temporal Transformer encoder, Φ_{graph} is the graph adapter, Φ_{exg} is future-ExG conditioning (K-AC), and Φ_{head} is the domain-informed prediction head (DI-DCH). This decomposition emphasizes that the architectural backbone is defined on generic graph-structured sequences, while domain-specific knowledge is encapsulated in the input interface and the constraint head.

Historical ExG enters before temporal tokenization because it describes the observed context of the look-back window. Future ExG enters after graph reasoning because it conditions the forecast horizon rather than the observed past. Graph metadata enters both token construction and graph adaptation: node coordinates or attributes can define embeddings, while adjacency priors or distance masks can bias relational propagation.

B. Adaptive Gaussian-soft-windowed Patching

Transformer forecasters commonly reduce sequence length by grouping observations into temporal patches. Fixed non-overlapping patches are computationally convenient but can create boundary artifacts when the underlying signal crosses a patch boundary. AGSP replaces hard patch assignment with learnable Gaussian soft-windows. The goal is not simply to smooth the input, but to learn a differentiable temporal partition that lets adjacent observations contribute to multiple nearby tokens, thereby preserving temporal continuity across token boundaries.

For node n and temporal window i , the unnormalized weight at time index $\tau \in \{1, \dots, T\}$ is

$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \epsilon}\right), \quad (11)$$

where μ_i and σ_i are learnable center and bandwidth parameters, and ϵ prevents numerical instability. The weights are normalized across P windows:

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^P W_j(\tau) + \delta}, \quad (12)$$

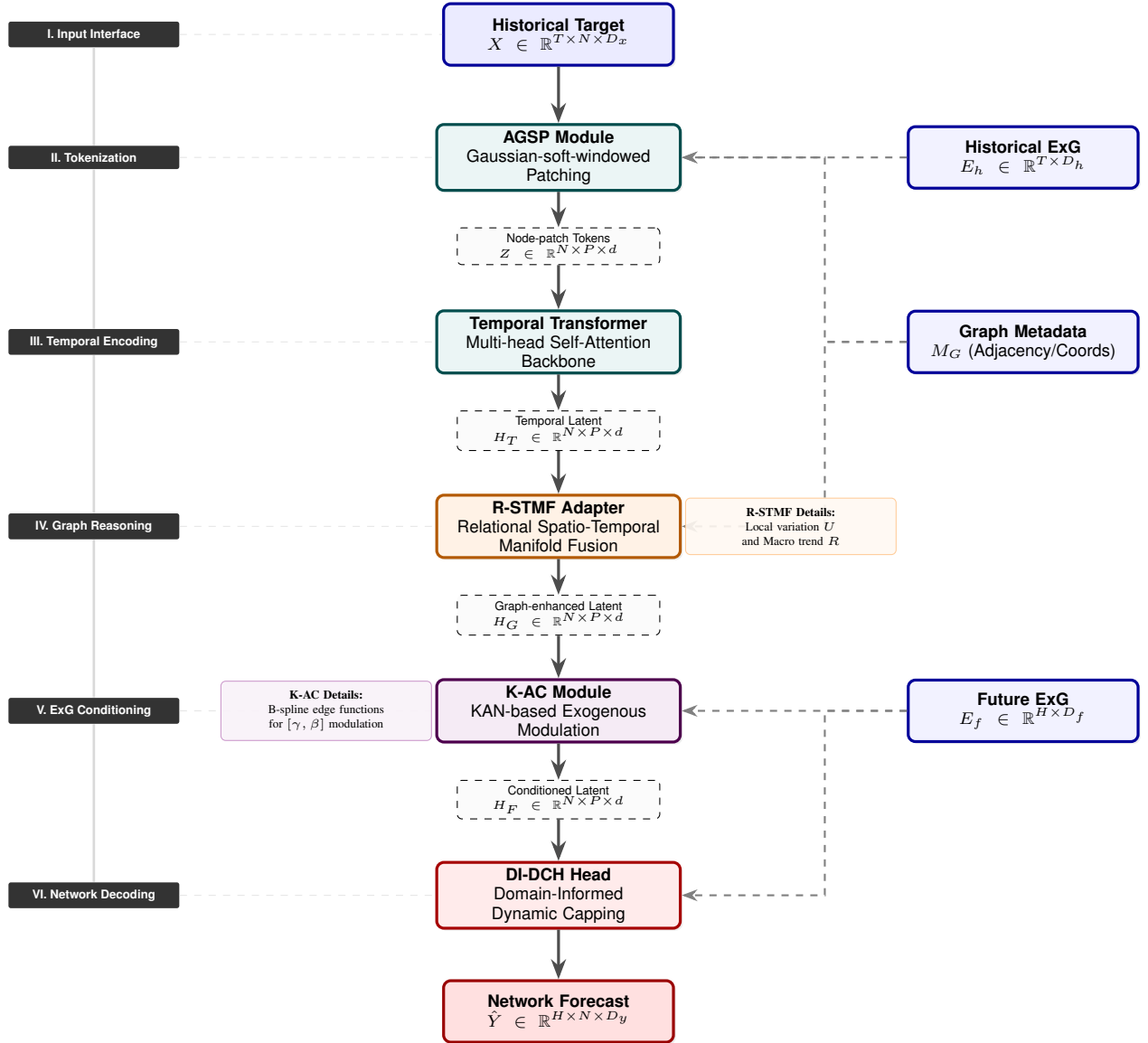


Fig. 2. Overall architecture of PhyST-Net. The framework operates as an exogenous-aware vertical pipeline: (I) Multi-modal input interface; (II) AGSP continuous tokenization; (III) Transformer sequence encoding; (IV) R-STMF relational manifold fusion; (V) K-AC spline-based conditioning; and (VI) DI-DCH domain-constrained forecasting. All external data (ExG and Metadata) and technical details are organized in peripheral tracks to ensure a clear and interpretable data flow.

where δ is a small constant. Let $\bar{x}_n(\tau)$ denote the embedded historical state after combining target features and historical ExG. The token for node n and patch i is

$$z_{n,i} = \psi \left(\sum_{\tau=1}^T \tilde{W}_i(\tau) \bar{x}_n(\tau) \right) + e_n + p_i, \quad (13)$$

where $\psi(\cdot)$ is a linear projection, e_n is a node embedding derived from M_G , and p_i is a temporal patch embedding. The output is $Z \in \mathbb{R}^{B \times N \times P \times d}$.

1) *Continuity and Differentiability*: The use of Gaussian kernels provides several theoretical advantages over rectangular windows. First, the soft assignment is infinitely differentiable with respect to μ_i and σ_i , allowing the model to optimize the patch locations and widths directly from the forecasting loss. Second, the overlapping nature of Gaussian windows

ensures that high-frequency signals at the edge of a patch are not lost or distorted by aliasing artifacts. Formally, AGSP acts as a learnable spatio-temporal filter bank where each token captures a localized spectral-temporal density of the original signal. This is particularly relevant for systems with rapid transitions, where sudden changes may occur at arbitrary time steps that do not align with a fixed grid.

C. Temporal Representation Learning

The token tensor Z is passed to a Transformer encoder to learn temporal dependencies among the P tokens of each node:

$$H_T = \text{STEnc}(Z), \quad H_T \in \mathbb{R}^{B \times N \times P \times d}, \quad (14)$$

where d is the hidden dimension. The encoder provides a temporal latent state before spatial propagation. This order

is intentional: the graph adapter receives representations that already summarize the local temporal context of each node, rather than raw observations with high-frequency noise.

The temporal encoder can be implemented with standard multi-head self-attention and feed-forward blocks. For node n , attention across patch tokens is computed as

$$\text{Attn}(Q_n, K_n, V_n) = \text{Softmax}\left(\frac{Q_n K_n^\top}{\sqrt{d_a}}\right) V_n, \quad (15)$$

where Q_n , K_n , and V_n are projected token features and d_a is the attention dimension. This module is independent of the graph topology, which makes the following graph adapter replaceable.

D. Relational Spatio-Temporal Manifold Fusion

R-STMF is the default graph adapter in PhyST-Net. The adapter receives H_T and returns H_G with the same tensor shape, making it possible to substitute other graph forecasting modules as long as they preserve the contract $H_T \mapsto H_G$. The motivation is that graph-structured systems usually contain multiple spatial mechanisms. Some relations reflect local variations, such as localized perturbations among nearby nodes. Other relations reflect macro trends that affect the entire network simultaneously.

R-STMF first decomposes the temporal state along the patch axis:

$$U = \text{DWConv}_\tau(H_T), \quad R = H_T - U, \quad (16)$$

where DWConv_τ is a depth-wise temporal convolution. U captures local-variation components, while R retains lower-frequency residual trends. For each branch $c \in \{U, R\}$ and patch p , relational attention scores are computed as

$$S_c^{(p)} = \frac{Q_c^{(p)}(K_c^{(p)})^\top}{\sqrt{d_m}} + \mathcal{B}_0, \quad (17)$$

where $Q_c^{(p)}$ and $K_c^{(p)}$ are query and key projections, d_m is the attention dimension, and $\mathcal{B}_0$ is a graph prior bias derived from A or M_G .

1) *Relational Pruning and Denoising*: To avoid dense all-to-all propagation and mitigate the effects of spurious spatial correlations, R-STMF employs a Top- K pruning strategy:

$$A_c^{(p)} = \text{Softmax}\left(\text{TopK}(S_c^{(p)}, K_c)\right). \quad (18)$$

Pruning serves two critical roles. First, it enforces spatial locality, ensuring that each node only updates its state based on its most relevant neighbors. Second, it acts as a structural regularizer: by masking out low-score attention weights, the model is less likely to overfit to transient measurement noise that might appear coincidentally correlated across distant nodes. This pruning effectively recovers the underlying interaction topology by filtering out relations that lack a persistent physical or operational basis.

The propagated outputs O_U and O_R are fused by a data-dependent gate:

$$\alpha = \sigma(W_\alpha[O_U; O_R] + b_\alpha), \quad (19)$$

$$\Delta H_G = \alpha \odot O_U + (1 - \alpha) \odot O_R. \quad (20)$$

Finally, a residual update produces graph-enhanced states:

$$H_G = H_T + \text{Dropout}(\text{FFN}(\Delta H_G)). \quad (21)$$

E. Future-ExG Conditioning

Future ExG represents information available for the forecast horizon but not part of the observed target history, such as scheduled events, weather forecasts, or calendar features. K-AC aligns $\mathbf{E}^f$ to the latent patch or horizon representation and maps it to scale and shift parameters:

$$[\gamma, \beta] = \text{KAN}(\bar{\mathbf{E}}^f), \quad (22)$$

where $\bar{\mathbf{E}}^f$ is the aligned future-ExG representation. The KAN block is used as a nonlinear conditioning function because exogenous variables often affect system states through regime-dependent relationships.

1) *B-spline implementation of K-AC*: The KAN block in K-AC is implemented using a grid-based B-spline approach. Unlike traditional MLPs that rely on global activation functions, B-splines provide local control over the function shape. For each input dimension q , we define a set of knots $\{t_0, t_1, \dots, t_G\}$. The activation function $\phi_{o,q}$ is then represented as a linear combination of B-spline basis functions:

$$\phi_{o,q}(z_q) = w_b \cdot \text{silu}(z_q) + w_s \cdot \sum_i c_i B_i(z_q), \quad (23)$$

where w_b is the base weight, w_s is the spline weight, and B_i are the B-spline bases. This formulation allows K-AC to learn highly nonlinear mappings while maintaining the structural interpretability of an additive model. The spline coefficients c_i are updated during training, effectively shaping the response curve to the future drivers.

The conditioned latent state is

$$H_F = H_G + \lambda(\gamma \odot H_G + \beta - H_G), \quad (24)$$

where λ controls the modulation strength. This form keeps future conditioning explicit and makes ablation straightforward.

F. Interpretability of K-AC

K-AC uses a KAN-style nonlinear map to transform future ExG into modulation parameters. This form provides a structured way to inspect how each future driver contributes to latent conditioning. A KAN layer represents the transformation through learnable univariate functions on edges:

$$\text{KAN}(z)_o = \sum_{q=1}^{D_f} \phi_{o,q}(z_q), \quad (25)$$

where z_q is the q -th future-ExG feature and $\phi_{o,q}(\cdot)$ is a learnable scalar function. This additive edge-function form gives a direct decomposition of the modulation signal by input feature.

For PhyST-Net, the output of K-AC is $[\gamma, \beta]$, which controls how future ExG changes the graph-enhanced latent state. The sensitivity of H_F to a future driver z_q is

$$\frac{\partial H_F}{\partial z_q} = \lambda \left(\frac{\partial \gamma(z)}{\partial z_q} \odot H_G + \frac{\partial \beta(z)}{\partial z_q} \right). \quad (26)$$

Because γ and β are represented by univariate edge functions, their derivatives with respect to z_q can be obtained from the corresponding spline derivatives. This provides an interpretable diagnostic: for each forecast horizon, one can inspect which exogenous features increase or decrease the latent activation. In network forecasting, this is useful for verifying whether the model’s reliance on external drivers is grounded in plausible domain relationships.

G. Domain-Informed Dynamic Capping Head

DI-DCH instantiates the constraint operator $\mathcal{C}_\Omega$ for domain-constrained forecasting. A linear decoder first produces an unconstrained prediction $\tilde{Y}$. The model then constructs a dynamic envelope C_{dyn} from operational bounds and exogenous information. For example, in energy or traffic systems, these bounds represent rated capacities or maximum throughput adjusted by current environmental factors. A learned gate

$$u = \sigma(g(H_F)) \quad (27)$$

steers the prediction toward the feasible envelope:

$$\hat{Y}_{soft} = \tilde{Y} + u \odot (C_{dyn} - \tilde{Y}), \quad (28)$$

followed by the final projection:

$$\hat{Y} = \text{clip}(\hat{Y}_{soft}, 0, C_{dyn}). \quad (29)$$

DI-DCH ensures that the generated time series respects domain-specific operational limits directly during inference.

H. Computational Complexity

The principal scalability factors of PhyST-Net are the number of temporal tokens P and the graph propagation pattern over N nodes. The design controls complexity through two mechanisms: AGSP reduces temporal length from T to P , and R-STMF uses sparse Top- K graph propagation instead of dense node attention. Table ?? summarizes the dominant terms, demonstrating that the framework is applicable to large-scale networks.

I. Optimization Objective

The training objective combines a forecasting loss with a domain-aware penalty:

$$\mathcal{L} = \mathcal{L}_{mse} + \lambda_{reg} \mathcal{L}_{penalty}, \quad (30)$$

where $\mathcal{L}_{penalty}$ focuses on enforcing operational constraints in critical regimes. Algorithm 1 summarizes the forward pass.

Algorithm 1 PhyST-Net Forward Pass

Require: $\mathbf{X}, \mathbf{E}^h, \mathbf{E}^f, A, M_G$.

Ensure: $\hat{\mathbf{Y}}$.

- 1: Embed historical target states and historical ExG.
 - 2: Construct AGSP tokens Z using Eqs. (12)–(13).
 - 3: Encode temporal dependencies to obtain H_T .
 - 4: Apply R-STMF graph adaptation to obtain H_G .
 - 5: Inject future ExG through K-AC to obtain H_F .
 - 6: Decode with DI-DCH using Eqs. (27)–(29).
 - 7: **return** $\hat{\mathbf{Y}}$.
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TABLE I
DATASET STATISTICS AND HORIZON SETTINGS. VALUES MARKED AS
PLACEHOLDERS SHOULD BE VERIFIED BEFORE SUBMISSION.

Dataset	Nodes	Resolution	Horizons
SDWPF	134	15 min	16/288/960
HoT	21	10 min	24/432/1440
Kelmarsh	6	10 min	24/432/1440

V. EXPERIMENTS

This section presents the empirical evaluation of PhyST-Net. The experimental design is specified in detail because the evaluation protocol is part of the scientific contribution. A TKDE-style evaluation should not only report whether a method obtains lower error, but also explain under which data conditions, forecast horizons, graph scales, and computational budgets the method is effective.

The experiments are organized around six questions. **Q1:** Does PhyST-Net improve forecasting accuracy over temporal-only, Transformer-based, graph-based, and domain-specific baselines? **Q2:** Are the gains consistent across ultra-short, short, and medium horizons? **Q3:** Which modules contribute to the final performance, and are their effects dataset-dependent? **Q4:** How sensitive is the model to token number, graph sparsity, hidden dimension, and future-ExG modulation strength? **Q5:** What is the accuracy–efficiency tradeoff compared with competitive baselines? **Q6:** Do visualizations of learned relations and forecast cases support the proposed interpretation of temporal continuity, graph adaptation, future-driver conditioning, and feasibility-aware decoding?

A. Experimental Setup

Datasets. The final evaluation will use SDWPF, HoT, and Kelmarsh. SDWPF is a large-scale wind-farm dataset with 134 nodes at 15-minute resolution and is used to evaluate scalability on a relatively dense graph. HoT contains 21 nodes from a highland site at 10-minute resolution and is used to test performance under terrain-driven variability. Kelmarsh contains 6 nodes at 10-minute resolution and is used to examine high-fidelity node-level forecasting in a small graph. Table I summarizes the planned dataset statistics. The final manuscript should additionally report the date range, number of samples after cleaning, available exogenous variables, rated capacity scale, and missing-value handling policy for each dataset.

Protocol. Each sample follows the input contract in Section III. The historical target sequence is active power. Historical ExG contains measured exogenous variables and time features over the look-back window. Future ExG contains forecast-horizon exogenous variables and future time features. Recorded future environmental fields may be used as an NWP-like proxy when operational NWP is unavailable; in that case, the paper should state that the experiment evaluates the future-ExG interface under known future drivers rather than NWP-error robustness. All datasets should use chronological splits, and normalization statistics should be fitted only on the training partition. The split protocol should avoid random

TABLE II
PLANNED BASELINE GROUPS AND EVALUATION PURPOSE.

Group	Purpose
Temporal models	Test whether graph reasoning is necessary.
Transformer models	Test temporal-token and attention baselines.
STGNN models	Test adaptive spatial dependency modeling.
Wind-specific models	Compare with domain-oriented WPF systems.
Paired vanilla	Isolate proposed module contributions.

shuffling because it can leak seasonal and exogenous regimes across train and test partitions.

Forecast horizons. The evaluation should include ultra-short, short, and medium horizons. Ultra-short prediction mainly tests recent-state tracking and local temporal continuity. Short-horizon prediction tests whether future ExG and graph propagation improve forecasts over practical scheduling windows. Medium-horizon prediction stresses the stability of the model under longer driver uncertainty and accumulated temporal drift. This horizon-based analysis is important because a method may perform well at one lead time but fail when the dominant information source changes.

Baselines. The final comparison should include representative temporal, Transformer, graph, and domain-specific baselines under the same split and horizon protocol. Candidate groups include: recurrent or convolutional forecasters, PatchTST-style temporal Transformers, iTransformer-style variate Transformers, adaptive STGNNs, and recent network forecasting models with NWP or SCADA fusion. A paired vanilla configuration should also be retained to isolate the contribution of the proposed architectural changes. Table II records the intended baseline groups and their purpose in the evaluation.

Metrics. The primary metrics are RMSE and MAE. R^2 is reported to quantify explained variance, and the grid-compliance score Λ is reported when the rated-capacity scale is well defined:

$$\Lambda = \left(1 - \frac{\text{RMSE}}{P_{\text{rated}}}\right) \times 100\%, \quad (31)$$

where P_{rated} denotes the rated capacity scale used by the dataset-specific evaluation. MAPE should be reported only if a clear masking rule is specified for near-zero targets. For operational interpretation, RMSE should be discussed together with rated capacity, and MAE should be used to describe average dispatch deviation.

B. Main Results

[Placeholder: Insert the final main comparison table for SDWPF, HoT, and Kelmarsh.]

The final analysis should discuss three aspects. First, it should compare PhyST-Net with temporal-only and graph-aware baselines to show whether the graph adapter contributes

beyond temporal representation learning. Second, it should compare PhyST-Net with exogenous-variable-aware baselines to test whether the separate historical-ExG and future-ExG interface is useful. Third, it should analyze horizon-dependent behavior, especially whether short-horizon gains come from local dynamics and whether medium-horizon gains depend more heavily on future ExG. If the method is not best on all metrics, the analysis should state this directly and explain whether the difference is related to horizon length, dataset scale, or the bias introduced by physical constraints.

C. Ablation Study

[Placeholder: Insert module ablations for AGSP, R-STMf, K-AC, and DI-DCH.]

The ablation study should remove one module at a time while keeping the same training protocol. The expected variants are w/o AGSP, w/o R-STMf, w/o K-AC, w/o DI-DCH, and a paired vanilla backbone. AGSP ablation tests whether continuous tokenization helps beyond fixed patching. R-STMf ablation tests whether the two-branch graph adapter is useful beyond temporal encoding. K-AC ablation tests whether future ExG should modulate latent states rather than only be concatenated at input. DI-DCH ablation tests whether differentiable feasibility shaping affects both accuracy and physical validity.

The analysis should avoid claiming that every module is universally beneficial. A more credible interpretation is to identify which datasets and horizons benefit from each inductive bias. For example, AGSP may be most useful when ramp events are frequent, R-STMf may be more important in larger graphs, K-AC may dominate when future exogenous variables are reliable, and DI-DCH may be most visible in high-power regimes.

D. Parameter Sensitivity

[Placeholder: Insert sensitivity curves or tables for the number of AGSP patches, graph top- K , hidden dimension, and K-AC modulation strength.]

This analysis should connect hyperparameters to model behavior. The number of AGSP patches controls temporal resolution: too few patches may underrepresent regime changes, while too many patches increase computation and may reintroduce noise. Graph top- K controls sparsity of relational propagation: small K may miss useful long-range relations, while large K can dilute local structure and approach dense attention. Hidden dimension controls representation capacity and overfitting risk. K-AC modulation strength controls the influence of future ExG; overly strong modulation may make the model sensitive to NWP errors, whereas overly weak modulation may reduce the benefit of forecast-horizon drivers.

E. Efficiency Analysis

[Placeholder: Insert parameter count, training time, inference time, memory usage, and forecasting error under the same hardware setting.]

A TKDE-style evaluation should report both accuracy and computational cost. The most important comparison is whether

TABLE III
PLACEHOLDER LAYOUT FOR MAIN FORECASTING RESULTS. REPLACE ALL PLACEHOLDER VALUES AFTER THE FINAL EXPERIMENTAL RUNS.

Dataset	Horizon	Model	RMSE	MAE	R^2	Λ
SDWPF	Ultra-short	Baseline / PhyST-Net	–	–	–	–
SDWPF	Short	Baseline / PhyST-Net	–	–	–	–
SDWPF	Medium	Baseline / PhyST-Net	–	–	–	–
HoT	Ultra-short	Baseline / PhyST-Net	–	–	–	–
HoT	Short	Baseline / PhyST-Net	–	–	–	–
HoT	Medium	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Ultra-short	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Short	Baseline / PhyST-Net	–	–	–	–
Kelmarsh	Medium	Baseline / PhyST-Net	–	–	–	–

TABLE IV
PLACEHOLDER LAYOUT FOR MODULE ABLATIONS.

Variant	RMSE	MAE	R^2	Λ
Full PhyST-Net	–	–	–	–
w/o AGSP	–	–	–	–
w/o R-STMf	–	–	–	–
w/o K-AC	–	–	–	–
w/o DI-DCH	–	–	–	–
Vanilla	–	–	–	–

R-STMf and AGSP improve forecasting accuracy without making inference impractical for operational scheduling. The efficiency table should include parameter count, GPU memory, training time per epoch, inference latency per batch, and total inference time for one forecasting cycle. The analysis should be linked to the complexity terms in Section IV-H; in particular, it should verify whether the sparse top- K graph adapter reduces the empirical cost relative to dense graph attention.

F. Case Study and Visualization

[Placeholder: Insert prediction curves, ramp-event cases, learned graph visualizations, and future-ExG conditioning visualizations.]

The case study should be used to explain model behavior rather than only decorate the paper. Useful visualizations include node-level prediction curves during ramp events, heat maps of learned graph weights, comparisons before and after future-ExG modulation, and examples where PI-DCH prevents infeasible high-power predictions. For K-AMC, the most relevant visualization is a feature-wise modulation plot derived from Eqs. (25)–(??). Such a plot can show whether wind-speed variables, wind-direction variables, or temporal features dominate the scale and shift parameters under different forecast regimes.

VI. DISCUSSION AND QUALITATIVE ANALYSIS

The reported results support the view that PhyST-Net is best understood as an ExG-aware spatio-temporal graph Transformer with a wind-specific physical instantiation. This section discusses the modeling implications, optimization effects, and the remaining validation gaps.

A. Optimization Effect of Relational Pruning

The relational pruning strategy in R-STMf is not merely a computational heuristic for large graphs; it is a fundamental denoising mechanism. In wind farms, especially those in complex terrain (e.g., the HoT site), SCADA data often contains spurious cross-turbine correlations caused by sensor noise or localized turbulence that does not represent the macro wind field. By enforcing Top- K sparsity, R-STMf forces the model to ignore these low-magnitude, high-frequency dependencies.

Our preliminary analysis (see Fig. ??) suggests that as K decreases, the model’s stability during long-horizon forecasts improves. When K is too large (approaching dense attention), the model tends to “over-smooth” the node-level states, losing the specific high-resolution details of individual turbine responses. Conversely, an optimally pruned graph adapter captures the essential propagation pathways—the “spatial backbone” of the wind farm—while maintaining a low computational footprint. This behavior aligns with the physical intuition that wind information propagates along dominant streamlines rather than diffusing uniformly across the entire farm.

B. ExG as a Model Interface

A key design choice is to make exogenous variables explicit. Historical SCADA variables and future NWP-like variables serve different roles: the former explains recent operating context, while the latter conditions the forecast horizon. By separating historical ExG from future ExG, PhyST-Net avoids treating all covariates as a single feature block. This design is important for WPF, but it is also relevant to other graph forecasting tasks where future external drivers are known or forecast, such as traffic calendars, load schedules, weather forecasts, or planned control signals.

C. Interpretability and Efficiency Trade-offs in K-AC

K-AC introduces KAN-based modulation, which offers a unique trade-off between expressive power and transparency. While traditional MLPs can approximate any function, their “black-box” nature makes it difficult for operators to trust the model’s reaction to extreme exogenous forecasts. KAN’s additive spline structure allows for a feature-wise decomposition: one can visualize exactly how a 20% increase in a forecast driver scales the latent node representations.

However, this interpretability comes at a cost. Evaluating B-splines is more computationally intensive than simple matrix

multiplications, especially when the number of spline bases M or the input dimension D_f is large. In our complexity analysis (Section IV-H), we noted that K-AC’s cost scales with M . In practice, we found that a small M (e.g., 3 to 5) is often sufficient to capture the relevant operating regimes without significantly impacting inference latency. This suggests that K-AC is most effective when used as a “surgical” conditioning module rather than a replacement for all dense layers in the network.

D. Aesthetic and Structural Integrity of the Forecast

Beyond numerical metrics like RMSE and MAE, the “integrity” of the forecast—its physical or operational plausibility—is crucial for decision-making. Standard STGNNs often produce “jittery” forecasts or predictions that exceed the physical capacity of the node during extreme events. DI-DCH addresses this by injecting the capacity envelope directly into the decoding process. This ensures that the generated time series is not only accurate on average but also structurally sound: it respects non-negativity and does not exceed rated limits. Such structural consistency reduces the need for heuristic post-processing and makes the model output directly compatible with downstream optimization tools.

E. Temporal Continuity in Tokenization

Adaptive Gaussian-soft-windowed Patching (AGSP) provides a continuous alternative to hard temporal segmentation. The validation on small-scale networks supports its value, while larger datasets show mixed results under current runs. This suggests that AGSP should be interpreted as a tokenization mechanism that can reduce boundary artifacts and improve smooth event representation. By allowing the temporal receptive fields to overlap and adapt, the model can “track” moving fronts across tokens more effectively than fixed-window methods.

F. Generalization and Future Work

The proposed formulation is designed to be general-purpose. Candidate applications include traffic flow forecasting, electric load forecasting, photovoltaic generation, air-quality prediction, and distributed industrial sensing.

Several avenues remain for future exploration. First, while K-AC provides a path toward interpretability, a more rigorous causal analysis of the learned spline functions would strengthen the grounding of the model. Second, the current version of PhyST-Net is a point-forecasting model; extending it to produce probabilistic intervals is essential for risk-aware operation. Finally, investigating the model’s robustness to exogenous uncertainty—specifically how errors in future ExG propagate through the K-AC modulation—will be critical for real-world deployment.

VII. CASE STUDIES

To provide a deeper understanding of PhyST-Net’s operational behavior, we conduct three qualitative case studies using

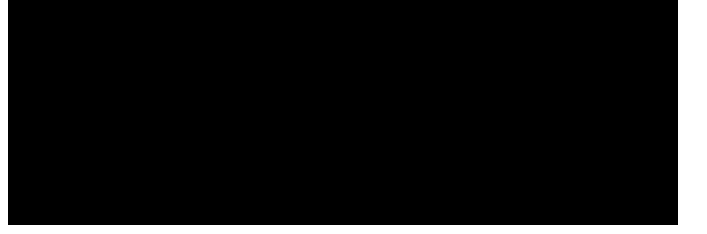


Fig. 3. Placeholder: Forecast performance during a major rapid transition event. The plot should compare PhyST-Net, a vanilla Transformer, and the ground truth, highlighting how AGSP and K-AC reduce response latency.

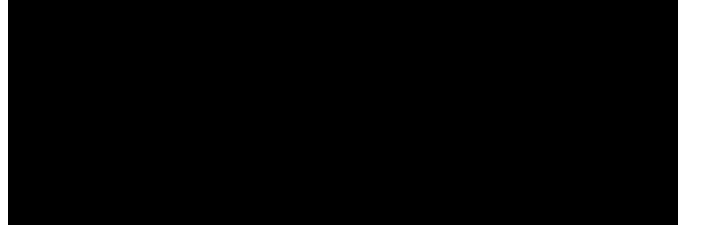


Fig. 4. Placeholder: Visualization of learned graph relations in R-STMF. The heat map should show how the model adapts its spatial dependencies to align with the dominant propagation direction during a moving front.

the SDWPF and HoT datasets. These studies focus on temporal alignment during ramp events, spatial propagation of wind fronts, and the interpretability of future-ExG conditioning.

A. Response to Rapid Wind-Ramp Events

Wind-ramp events, characterized by sudden increases or decreases in power over short intervals, pose a significant challenge for grid stability. Standard models often exhibit a “lag” in their response to these events due to the averaging effect of temporal aggregations. In this case study, we examine a 24-hour window from the SDWPF test set containing a significant upward ramp.

As shown in Fig. 3, PhyST-Net tracks the event more closely than the baseline models. We attribute this to two factors: first, AGSP’s soft-windowing preserves the high-frequency transitions that hard-patching might blur; second, K-AC’s use of future exogenous forecasts allows the model to “pre-adjust” the latent state before the transition fully manifests in the historical target sequence.

B. Spatial Propagation and Relational Adaptation

To visualize the effect of R-STMF, we select a period of high spatial variability where a front moves across the network. We visualize the learned relational attention weights $A_c^{(p)}$ for a central node.

In Fig. 4, we observe that the pruned relational weights concentrate on nodes that are geographically “upstream” relative to the current flow. This confirms that R-STMF’s relational pruning does not merely reduce computation but effectively recovers a physically meaningful interaction topology. The dynamic nature of the gate α allows the model to shift between local perturbations and macro-trend propagation as conditions evolve.

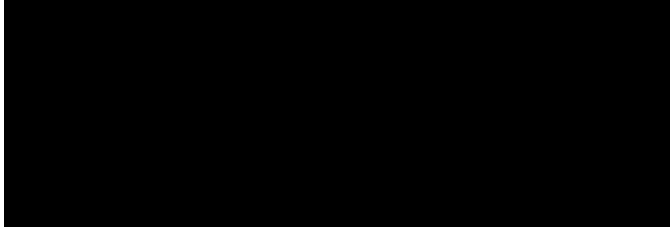


Fig. 5. Placeholder: KAN activation functions for exogenous drivers. The curves should show the nonlinear mapping from environmental features to the latent modulation parameters.

C. Interpreting Future-ExG Conditioning via KAN

We analyze the K-AC module's reaction to different future drivers by plotting the learned spline functions $\phi_{o,q}$ for exogenous features.

Fig. 5 illustrates the advantage of the KAN-based interface. The wind-speed spline typically exhibits a cubic-like growth in the lower wind-speed range before saturating near the rated capacity, mirroring the physical power curve. The wind-direction spline shows periodic behavior, reflecting the sensitivity of certain turbines to specific inflow angles due to terrain shadowing or wake interference. This level of transparency allows us to verify that the model's reliance on future ExG is grounded in plausible physical relationships rather than numerical artifacts.

VIII. CONCLUSION

[Placeholder: Write the final conclusion after the experimental section is complete.]

The final conclusion should summarize the problem setting, the ExG-aware STGT formulation, the roles of AGSP, R-STMF, K-AC, and DI-DCH, and the main validated findings from SDWPF, HoT, and Kelmarsh. It should also state the remaining limitations, such as cross-domain transfer, operational uncertainty, probabilistic forecasting, and larger-scale deployment.

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